Field-cooperative neural topic modelling of civil engineering literature: missing-abstract inference

Abstract

Engineering collections often contain records with titles but no abstracts. We study how to organise civil engineering title–abstract records into interpretable topics and infer topics when the abstract is missing. We present FC-NTM, a field-cooperative neural topic model that encodes each field in its own variational stream over one shared topic decoder; masking lets one trained model serve complete, partial and title-only records, and a unified topic–word protocol separates read-out from model effects. On 8,000 OpenAlex civil engineering records, FC-NTM attains the highest class TF–IDF coherence on the civil corpus; its generative β coherence is 0.584. In ten-seed paired comparisons it is ahead of BTM in all ten seeds on this corpus for class TF–IDF (+0.049), while the generative β difference is +0.004 (6/10, 95% CI $[-0.016, +0.023]$); the matched gain is +0.028 on GitHub (9/10) and −0.009 on AG News (3/10). A same-budget concatenated-field control adds +0.008–+0.037 coherence, while field-level posteriors and missing-field inference require the two-stream model. Title-only masking retains the topic of 67.3% of civil records (mean JS 0.033); under 10%–80% abstract masking, its mean posterior JS is the lowest of the six compared models at every rate, and topic retention is 94.5%–70.9% versus 82.9%–67.3% for BTM; title-only routing reaches 0.664 accuracy on 42.9% of test records covered by 15 OpenAlex topics, against 0.810 for a supervised classifier. Occlusion experiments place the clearest all-baseline stability advantage on GitHub issues; on the civil corpus and AG News, BTM or LDA degrade slightly less.

Keywords: Neural topic model, field cooperation, neural variational inference, missing data, civil engineering literature, text mining

1 Introduction

Paper and report collections in engineering libraries and design offices grow faster than anyone can read them, and in many catalogues the abstract is simply absent — the title is all that circulates. If the records could still be sorted into interpretable topics with titles alone, staff could browse and screen a collection cheaply, leaving the fine cataloguing to experts. Title–abstract records are one instance of a pattern that recurs across engineering documents: a short summary-like field sits next to a longer

contextual field, as in a clause and the contract around it, or a report headline and its body. The two fields rarely say the same thing in the same way. Bodies carry context and causes; summaries carry objects and actions.

Topic models reduce an unlabelled collection to a set of interpretable topic structures, and domain-specific text mining relies on them in settings as varied as financial news [10], software repositories and social media. Latent topics have also been coupled with downstream neural encoders for large-scale text classification [30]. The model families themselves have shifted over the past decade, from probabilistic graphical models to neural variational ones. Latent Dirichlet allocation (LDA) [4] is the classical bag-of-words Bayesian model. The Gibbs sampling Dirichlet–multinomial mixture model (GSDMM) targets short texts with collapsed Gibbs sampling [48]. The neural variational document model (NVDM) was an early application of variational auto-encoders to text [19]. Srivastava and Sutton introduce autoencoding variational inference for topic models (AVITM) and the ProdLDA model [36]. Contextualized topic models such as CombinedTM combine contextual document embeddings with bag-of-words counts [2]; the embedded topic model (ETM) places topics in a word-embedding space [7]. The biterm topic model (BTM) learns topics from corpus-level word-pair co-occurrences and provides a baseline for short texts [47]. Surveys treat structured and short-text corpora as open problems for neural topic models [46].

Concatenating fields into a single bag of words removes their explicit boundaries. Multi-view and multi-source models instead learn from distinct observations [13, 18, 26, 50]. Structured neural topic models encode user–item context and infer sentence-level aspects in reviews [8], while structural topic models incorporate document-level covariates [32]. Related short-text models address dual-view topic learning [18], reinforce topic inference with context in the context-reinforced neural topic model (CRNTM) [9], or learn dual-view representations for interpretable dynamic topic modelling (DVI-DTM) [17]. Kurle et al. condition separate encoders on different information sources [13]. Pergola et al. use paired reviews and plots to disentangle opinion and plot topics [26], and Esmaeili et al. use user–item context to structure review topics into inferred aspects [8]. In our setting, the views are fixed fields of the *same* record. Separate variational streams learn their relationship without requiring additional aspect labels. We apply this field-level configuration with one shared vocabulary and one linear decoder, and evaluate missing-background inference and field-restricted read-outs.

Engineering researchers use text mining to analyse safety and accident reports [39, 49] and to classify regulatory documents, detect contractual risk clauses and predict risk-handling actions [21, 27, 34, 53]. Domain surveys review these applications and identify directions for natural language processing in construction [35, 44]. Researchers also use topic models to analyse urban events with DUET [42], map machine-learning applications in buildings [54] and develop taxonomies for digital building passports [40]. Our application focuses on title–abstract records: topic inference when the abstract is missing, field-restricted read-out, and the reliability of comparisons between such read-outs.

We make three contributions.

1. **A unified topic–word read-out protocol for comparative evaluation.** All models are scored on one evaluation vocabulary with exactly ten words per topic. The read-out-matched comparison evaluates each included model through both its topic–word matrix and a class term-frequency–inverse-document-frequency (TF–IDF) read-out. The protocol separates differences caused by the read-out from differences caused by the model, a common confound in topic-model benchmarking, and makes the reported comparisons interpretable across models. We treat this protocol as the primary methodological contribution.
2. **A field-cooperative model as a reference implementation.** FC-NTM encodes the two fields of a record in separate conditional variational streams that share one linear topic decoder. Cross-field reconstruction and a field-consistency regulariser tie the two fields into a shared topic space, and field-level word masking with whole-field dropout lets a single trained model serve complete, partially observed and title-only records without retraining or architectural change.
3. **Multi-seed evidence, a single-encoder control and an engineering screening task.** Ten-seed paired comparisons show consistent class TF–IDF gains on the civil engineering corpus (+0.049; 10/10 seeds) and on GitHub issues (+0.028; 9/10 seeds); AG News shows no gain (−0.009; 3/10 seeds). The generative β difference is +0.004 on the civil corpus (6/10; 95% CI $[-0.016, +0.023]$), and negative on the two public corpora. A same-budget concatenated-field control identifies the boundary of the contribution: it adds +0.008–+0.037 coherence over a single encoder, while separate field posteriors and trained missing-field inference require the two-stream configuration. A 10%–100% abstract-masking curve on the civil corpus shows the lowest mean posterior JS at every rate and higher topic retention than BTM up to 80% masking. A title-only routing experiment on the civil corpus reaches 0.664 accuracy on the evaluated subset and shows how the trained model screens engineering records from titles alone.

Figure 1 summarises the problem setting and the three contributions: dual-field scholarly records whose abstracts may be missing, the field-cooperative configuration that trains both fields against one shared decoder, and the observation regimes a single trained model has to serve.

2 Related work

Neural topic models.

Many neural topic models based on variational autoencoders (VAEs) encode bag-of-words representations into continuous latent posteriors. NVDM was an early application of variational auto-encoders to text [19]; AVITM adapted this approach to LDA and introduced ProdLDA [36]. CombinedTM concatenates projected contextual document embeddings with bag-of-words counts [2], ETM places topics in an embedding space [7], and BTM models corpus-level word-pair co-occurrences for short texts [47]. BERTopic builds its topic representations from class TF–IDF [11]; Top2Vec derives topic vectors from a joint document–word embedding space [1]. GraphBTM and BCK-TM add word-pair structure [51, 56], and BigARTM contributes an additive regularisation framework [41]. Topic-distribution quantisation

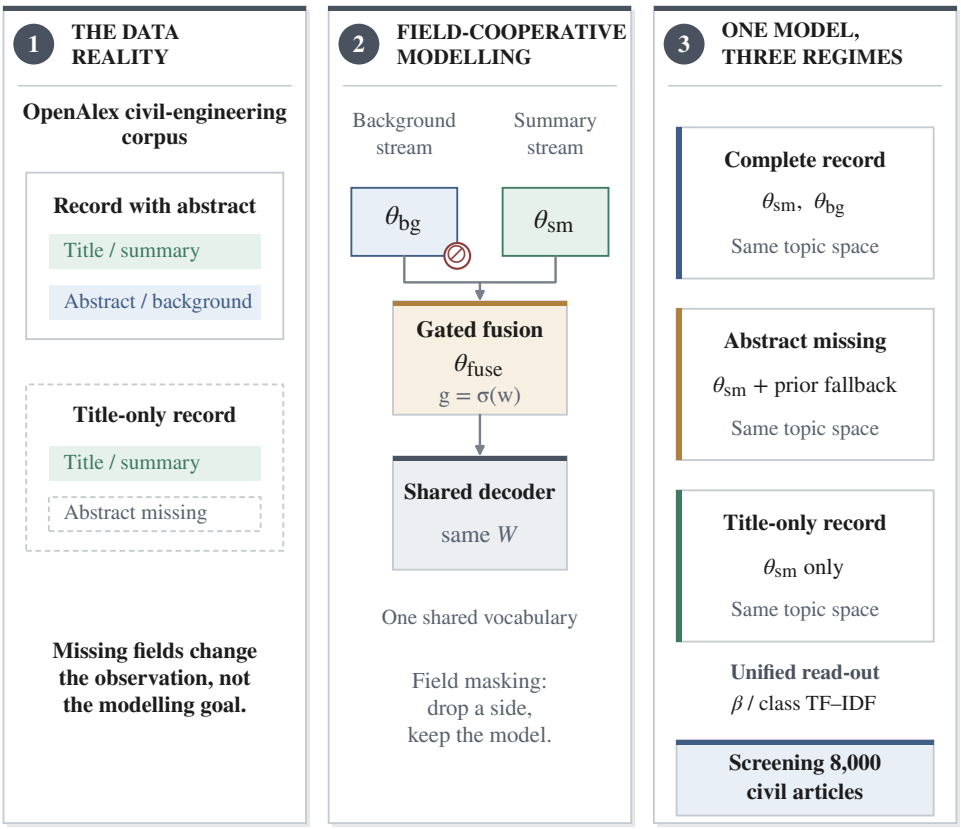


Fig. 1 Problem setting and contributions. (1) The application corpus is dual-field scholarly metadata in which the abstract (background field) may be missing, leaving title-only records. (2) FC-NTM encodes both fields with separate conditional variational streams tied to one shared topic–word decoder; field masking drops a side at training or inference time without changing the model. (3) One trained model serves complete, abstract-missing and title-only records through a unified topic–word read-out, supporting topic-based screening of civil-engineering literature

with a negative-sampling decoder is a more recent short-text option [45]. Our decoder follows the ProdLDA formulation. We apply a topic-orthogonality penalty, dropout [37] and validation-based early stopping [28].

Multi-view, multimodal and structured-document models.

Multi-view latent-variable models learn shared structure while retaining distinct observations. Kurle et al. use source-specific encoders and compare their posterior approximations [13]. Zhang et al. incorporate anomaly detection into robust multi-view topic modelling [50], while Pergola et al. separate opinion topics from plot topics [26] and Liu et al. study dual-view topic learning for short texts [18]. Structured review models use user–item context and sentence-level aspects [8];

metadata-conditioned models incorporate document metadata [6, 32]. CRNTM reinforces short-text topic inference with context [9], and DVI-DTM learns dual-view representations for dynamic topics [17]. Other short-text models exploit relational or sparse biterm structure [16, 55]. Suzuki et al. develop JMVAE-kl, a joint multimodal variational autoencoder variant, to support inference with missing modalities [38], and Nguyen et al. model topics at the sub-document level [23]. We apply these ideas at the field level: two textual fields, one shared vocabulary and one linear decoder, evaluated through missing-background inference and field-restricted read-outs.

Text mining and topic modelling of engineering documents.

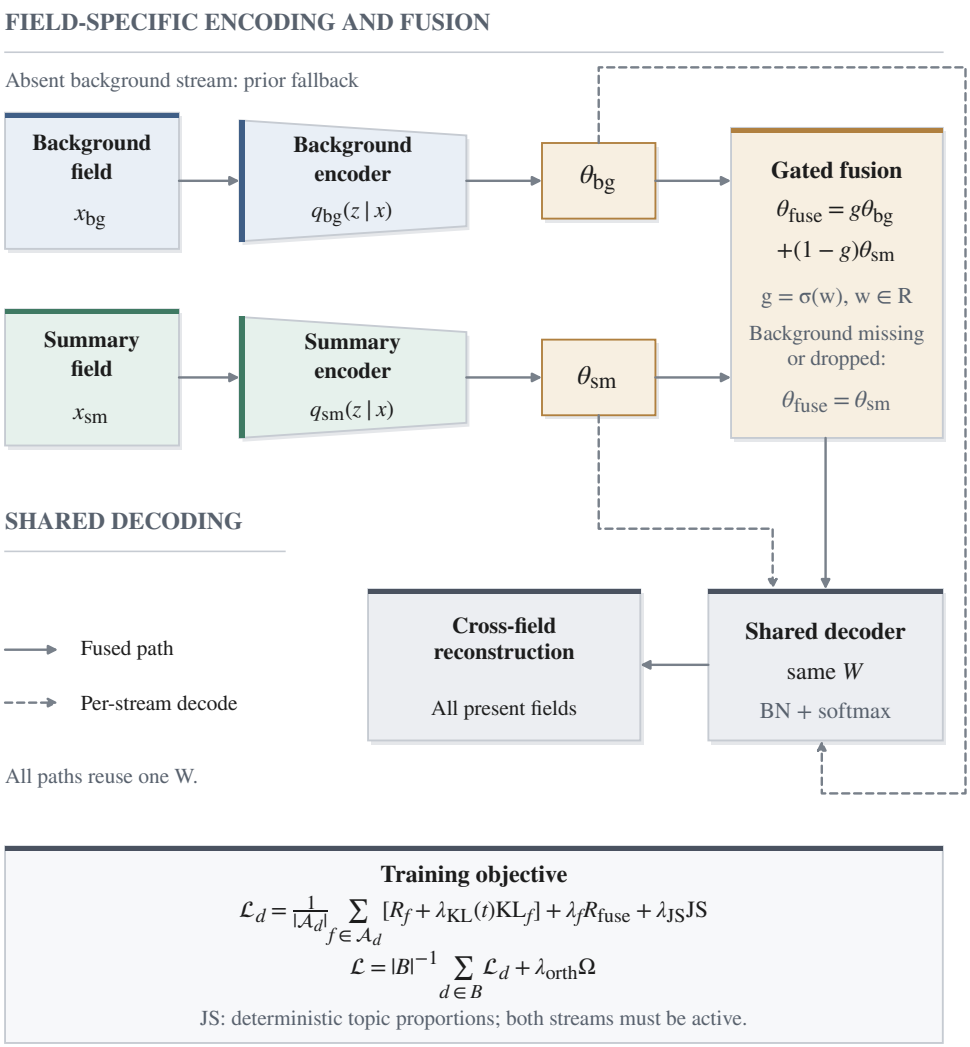
Tixier et al. extract precursors and outcomes from construction injury reports [39], and Zhang et al. analyse construction-site accidents with text mining and natural language processing [49]. Salama and El-Gohary classify construction text to support automated compliance checking [34]. Zhou and El-Gohary use ontology-based multilabel classification for regulatory documents [53]. Moon et al. detect contractual risk clauses with bidirectional encoder representations from transformers (BERT) [21], while Pham and Han predict risk-handling actions in construction contracts [27]. Domain surveys synthesise this work [35, 44]. For topic-modelling applications, Wang and Taylor propose DUET [42], Zhou et al. map machine-learning applications in buildings [54], and Vangelova et al. develop taxonomies for digital building passports [40]. We examine how fixed title–abstract fields affect field-restricted read-out and missing-abstract inference.

3 Methodology

Figure 2 gives an overview of FC-NTM. The two fields are encoded by separate conditional variational streams that share one topic decoder; the streams are tied by cross-field reconstruction and a Jensen–Shannon (JS) consistency term, and field-level masking with whole-field dropout covers partially and fully missing fields. The objective also includes Kullback–Leibler (KL) regularisation and a topic-orthogonality penalty.

3.1 Problem setting

Each record d consists of a *background* field (e.g. the abstract of a paper or the body of an issue) and a *summary* field (e.g. the title), with N records in total and a shared evaluation vocabulary of V terms. The model maintains K topics, a document–topic distribution $\theta_d \in \Delta^{K-1}$, and a topic–word matrix $\beta \in \mathbb{R}^{K \times V}$ shared by both fields. The task is to infer θ_d and β from unlabelled records, to produce readable topics under two read-outs, and to keep the model usable when one field is partially or entirely missing at inference time.



Training objective

$$\mathcal{L}_d = \frac{1}{|\mathcal{A}_d|} \sum_{f \in \mathcal{A}_d} [R_f + \lambda_{KL}(t)\mathrm{KL}_f] + \lambda_f R_{fuse} + \lambda_{JS}\mathrm{JS}$$

$$\mathcal{L} = |B|^{-1} \sum_{d \in B} \mathcal{L}_d + \lambda_{orth}\Omega$$

JS: deterministic topic proportions; both streams must be active.

Fig. 2 Overview of FC-NTM. Each field has its own conditional variational stream. Solid arrows show the fused path, in which the gated mixture is decoded and every present field is reconstructed; dashed arrows show the per-stream decodes, which reuse the same decoder parameters. The training objective combines per-stream reconstruction, the KL terms of the active streams, fused reconstruction, the JS term (computed only when both streams are active) and the topic-orthogonality penalty. A missing or dropped field falls back to the prior, so the gated mixture reduces to the remaining stream ($\theta^{\mathrm{fuse}} = \theta_{\mathrm{sm}}$ when the background field is missing or dropped). The β read-out uses the rows of W

3.2 Field streams and the shared decoder

Each field is encoded by its own two-layer multilayer perceptron (MLP) encoder (200 hidden units, Softplus activations, dropout rate 0.2) into the parameters of a field-conditioned Gaussian posterior,

$$q_f(z_f \mid x_f) = \mathcal{N}\big(\mu_f(x_f), \operatorname{diag}(\sigma_f^2(x_f))\big), \qquad f \in \{\mathrm{bg}, \mathrm{sm}\}, \tag{1}$$

where x_f is the L_1-normalised bag of words of field f and $\log \sigma_f^2$ is capped at 2. Posterior means are mapped to topic proportions by a deterministic softmax, $\bar{\theta}_f = \operatorname{softmax}(\mu_f(x_f))$, which replaces sampling at evaluation time. The deterministic softmax of the posterior mean gives a point estimate of the topic proportions. Both streams share one linear topic–word matrix $W \in \mathbb{R}^{K \times V}$, and we write $\theta_f \in \mathbb{R}^{1 \times K}$ for the topic proportions of field f under the row-vector convention,

$$p(w \mid \theta_f) = \operatorname{softmax}\big(\mathrm{BN}(\theta_f W)\big), \tag{2}$$

where θ_f passes through dropout before the product, BN is a batch-normalisation layer *without* affine parameters applied over the vocabulary dimension (batch statistics during training, running statistics at evaluation), and there is no additive bias. The β read-out of Section 3.6 reads the rows of W; generation follows the normalised path of Eq. (2).

3.3 Training objective

For a document d let $\mathcal{G}_d$ collect its present fields (the background field is present when the record carries it; the summary stream is always instantiated, including zero-vector inputs after preprocessing) and let $\mathcal{A}_d \subseteq \mathcal{G}_d$ be the *active streams*, i.e. present fields whose stream is not removed by whole-field dropout. Write $n_{dw}^{(g)}$ for the raw count of word w in field g. Each active stream decodes once (Eq. (2)) and reconstructs *every present field*,

$$R_f = \sum_{g \in \mathcal{G}_d} \omega_g \, R_f(g), \qquad R_f(g) = -\sum_{w=1}^{V} n_{dw}^{(g)} \log p(w \mid \theta_f)\,, \tag{3}$$

with fixed field weights $\omega_{\mathrm{bg}} = 2.0$ and $\omega_{\mathrm{sm}} = 1.0$. The summary stream also reconstructs the background field (cross-field reconstruction). A background field that is observed but dropped during training remains a reconstruction target even though its own stream is inactive, whereas a naturally missing background field enters neither the inputs nor the targets. The per-document objective is

$$\mathcal{L}_d = \frac{1}{|\mathcal{A}_d|} \sum_{f \in \mathcal{A}_d} \Big[R_f + \lambda_{\mathrm{KL}}(t)\, \mathrm{KL}_f \Big] + \lambda_f R_{\mathrm{fuse}} + \lambda_{\mathrm{JS}}\, \mathrm{JS}, \tag{4}$$

where $\mathrm{KL}_f = \mathrm{KL}\big(q_f(z_f \mid x_f) \,\|\, \mathcal{N}(0, I)\big)$ is summed over the K latent dimensions and clamped at zero, and $\lambda_{\mathrm{KL}}(t) = 0.05 \min(1, t/20)$ anneals linearly during the first 20

epochs. The fused term $R_{\text{fuse}} = \sum_{g \in \mathcal{G}_d} \omega_g R_{\text{fuse}}(g)$ is a second reconstruction of both present fields from the fused topic mixture

$$\theta_d^{\text{fuse}} = g\,\theta_{\text{bg}} + (1-g)\,\theta_{\text{sm}}, \qquad g = \sigma(w), \quad w \in \mathbb{R}. \tag{5}$$

for records with both fields present, and $\theta_d^{\text{fuse}} = \theta_{\text{sm}}$ when the background field is missing or dropped. During training Eq. (5) combines the *sampled* per-stream proportions that also feed the per-stream reconstructions; at inference time, and for the JS term below, the deterministic estimates $\bar{\theta}_f$ are used, so that Eq. (5) becomes a convex combination of two point estimates. We refer to this quantity as a *fused topic mixture* to distinguish it from a posterior. The gate g receives gradients only through R_{fuse}, so with $\lambda_f = 0$ the model degenerates exactly to fixed mean fusion. The consistency term

$$\text{JS} = \tfrac{1}{2}\,\text{KL}\big(\bar{\theta}_{\text{bg}} \| m\big) + \tfrac{1}{2}\,\text{KL}\big(\bar{\theta}_{\text{sm}} \| m\big), \qquad m = \tfrac{1}{2}\big(\bar{\theta}_{\text{bg}} + \bar{\theta}_{\text{sm}}\big), \tag{6}$$

acts on the deterministic topic proportions and does not use the Gaussian posteriors. It is computed only for documents whose two streams are both active and is not applied to records with one active stream. A topic-orthogonality penalty keeps the rows of W apart,

$$\Omega = \frac{1}{K(K-1)} \sum_{k \neq k'} \big|\cos\langle W_k, W_{k'}\rangle\big|, \tag{7}$$

and the batch objective is $\frac{1}{|B|}\sum_{d \in B} \mathcal{L}_d + \lambda_{\text{orth}}\Omega$ with $\lambda_{\text{orth}} = 0.2$.

3.4 Field masking

Two masking mechanisms expose the model to missing fields, and they act on different units: word types during training, token positions at test time (Section 5.2). (i) *Word-level input dropout* (denoising) is applied during training to the bag-of-words *word types* of each field: each bag-of-words dimension is independently zeroed with probability 0.30 (background) or 0.40 (summary) in the encoder view only, while the reconstruction targets remain the complete counts. (ii) *Whole-field dropout* removes the background stream of each training document with probability 0.5 and substitutes the prior; the summary stream is never dropped and learns to reconstruct both fields on its own. The consistency weight $\lambda_{\text{JS}} = 0.5$, the fusion weight $\lambda_f = 0.5$ and the dropout probabilities were selected on held-out data.

3.5 Optimisation

Training uses Adam (learning rate 5×10^{-4}; exponential decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.999$), mini-batches of 128, gradient-norm clipping at 5.0 and a budget of 700 epochs. After the KL warm-up the state with the lowest held-out loss is kept, for which 20% of the training split is held out; the patience is 150 epochs. Evaluation-mode inference is deterministic: the reparameterisation sampling is switched off and the held-out loss uses posterior means. Numerical-anomaly recovery rolls the weights back to the most recent finite epoch, clears the optimiser moments and halves the

learning rate; this recovery step is separate from the best-held-out-epoch selection of the early-stopping loop.

3.6 Unified topic–word read-out

We fix one evaluation vocabulary and exactly ten words per topic; every model is read out under this protocol. The β **read-out** takes the top ten words of each normalised row of the model's topic–word matrix; for FC-NTM these are the raw rows of W, whereas generation uses Eq. (2). LDA, BTM, NMF, ProdLDA, ETM, CTM-lite, CombinedTM and FC-NTM are all evaluated this way. The **class TF–IDF read-out** (abbreviated cTFIDF in the tables; figures with built-in labels use the short form cTF) aggregates the training-term counts by the soft document–topic assignment of the model, applies a BERTopic-style class TF–IDF transform [11], and then takes the top ten words. Soft assignments replace the hard cluster membership used in the standard class-level aggregation; for FC-NTM this read-out is built from the fused topic mixture $\bar{\theta}^{\text{fuse}}$, which is why the two read-outs can rank models differently.

4 Experimental setup

4.1 Corpora and evaluation protocol

We use three dual-field corpora. The **civil engineering corpus** is the application corpus: title–abstract records of English-language scholarly works from OpenAlex [29] (https://openalex.org) in *Civil and Structural Engineering* and *Building and Construction* (2000–2023; CC BY, CC BY-SA, or public-domain records). The sample comprises 7,102 articles, 889 conference papers and 9 reviews, split into 6,000 training and 2,000 test records, with the title as the summary field and the abstract as the background field (mean length ratio 11.2). **GitHub issues** [25] pairs the issue body with its title (four mature repositories, 6,000/2,000, ratio 12.9), and **AG News** [52] pairs the description with the headline (6,000/7,600, ratio 3.9). All corpora are lowercased and tokenised on letter sequences; stop words and very short tokens are removed, the vocabulary is built from the training split only, and test records never enter the vocabulary. Tables 1 and 2 report the two public corpora and Table 3 the application corpus, all on the same evaluation vocabulary with exactly ten words per topic and $K = 30$. The evaluation protocol (vocabulary, ten words per topic, coherence and diversity metrics) is shared by every model, whereas the training procedure differs across algorithm families and is described in Section 4.2. Public-corpus results are also reported on development-free subsets (1,600 GitHub and 7,200 AG News records). The results are organised around three questions: how well civil title–abstract records can be structured into verifiable topics; how much signal remains when the abstract is missing; and how much the complete field-cooperative configuration adds over a concatenated single-encoder control. Cross-corpus diagnostics complete the analysis.

4.2 Baselines, training protocols and metrics

Baselines are LDA [4], BTM [47], non-negative matrix factorisation (NMF) with a KL objective [15], ProdLDA [36], ETM [7], our lightweight contextual baseline

Table 1 Topic quality on AG News under the unified evaluation vocabulary (ten words per topic; values are means over seeds 11/23/37 where applicable; BERTopic and Top2Vec choose their own topic numbers and are shown for reference). The FC-NTM β row uses the generative softmax(BN($e_k W$)) read-out; the raw decoder view is in Online Resource 1 (Table S5).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.600±0.011	0.164±0.005	−4.19	0.409	0.980
CombinedTM	30	0.526±0.037	0.057±0.070	−8.01	0.118	1.000
FC-NTM-cTFIDF	30	**0.612**±0.014	**0.173**±0.012	−4.20	0.193	0.974
FC-NTM	30	0.541±0.041	0.100±0.047	−8.13	0.851	0.999
ETM	30	0.506±0.014	0.092±0.010	−5.12	0.439	0.982
ProdLDA	30	0.526±0.031	0.071±0.015	−6.61	0.055	0.999
Top2Vec	2.7	0.498±0.032	0.099±0.031	−2.92	0.268	0.937
NMF	30	0.494±0.015	0.069±0.013	−5.96	0.799	0.997
BERTopic	29	0.488±0.022	0.035±0.026	−8.81	0.681	0.997
CTM-lite	30	0.431±0.023	−0.015±0.016	−8.34	0.041	0.998
LDA	30	0.421±0.013	−0.066±0.018	−8.62	0.696	0.994

Table 2 Topic quality on GitHub issues (same evaluation protocol and read-outs as Table 1; values are means over seeds 11/23/37; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.712±0.012	0.232±0.012	−1.79	0.349	0.934
FC-NTM-cTFIDF	30	**0.727**±0.022	**0.274**±0.026	−1.90	0.133	0.901
FC-NTM	30	0.625±0.026	0.192±0.038	−5.10	0.552	0.988
ETM	30	0.675±0.014	0.213±0.010	−2.14	0.366	0.955
CombinedTM	30	0.643±0.022	0.193±0.015	−4.70	0.031	0.983
NMF	30	0.614±0.005	0.164±0.003	−3.26	0.595	0.985
ProdLDA	30	0.579±0.007	0.107±0.012	−5.21	0.043	0.995
CTM-lite	30	0.585±0.032	0.129±0.036	−5.11	0.011	0.989
LDA	30	0.546±0.011	0.074±0.017	−4.21	0.684	0.990
BERTopic	29	0.540±0.015	0.076±0.012	−7.00	0.546	0.976
Top2Vec	160	0.493±0.009	−0.008±0.013	−9.65	0.337	0.994

(CTM-lite), the official CombinedTM [2], BERTopic [11] and Top2Vec [1]; contextual models use Sentence-BERT embeddings [31]. The implementations use different training procedures. BTM uses collapsed Gibbs sampling, whereas LDA uses the variational inference implementation in Gensim. NMF uses matrix factorisation. Among the neural variational models, ProdLDA, ETM, CTM-lite and FC-NTM share the Adam loop [12] and the 700-epoch budget but use different learning rates; the official CombinedTM retains its own training loop. BERTopic and Top2Vec use embedding-based clustering and choose their own topic numbers; they are reported as reference systems. Full hyper-parameters are listed in Online Resource 1. Coherence measures

Table 3 Topic quality on the civil engineering corpus under the unified evaluation vocabulary (ten words per topic; values are means over seeds 11/23/37; BERTopic and Top2Vec choose their own topic numbers and are shown for reference). For FC-NTM, the β row uses the generative softmax(BN($e_k W$)) read-out; the raw decoder view is reported in Online Resource 1 (Table S5).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.583±0.010	0.128±0.009	−2.18	0.283	0.944
CombinedTM	30	0.539±0.029	0.082±0.028	−6.11	0.063	0.998
FC-NTM-cTFIDF	30	**0.636**±0.014	**0.159**±0.011	−2.21	0.165	0.937
FC-NTM	30	0.584±0.029	0.118±0.028	−5.41	0.744	0.998
ETM	30	0.581±0.012	0.119±0.007	−2.75	0.379	0.963
ProdLDA	30	0.580±0.023	0.125±0.017	−3.84	0.051	0.999
Top2Vec	153	0.515±0.002	0.029±0.011	−8.52	0.359	0.995
NMF	30	0.598±0.009	0.135±0.006	−2.86	0.559	0.984
BERTopic	29	0.576±0.026	0.116±0.027	−5.06	0.563	0.991
CTM-lite	30	0.536±0.030	0.098±0.027	−3.85	0.013	0.995
LDA	30	0.554±0.008	0.056±0.025	−4.62	0.602	0.987

assess topic interpretability through word co-occurrence [22]; we report C_V [33], normalised pointwise mutual information (NPMI) [5, 14] and UMass [20]. We report inverted rank-biased overlap (IRBO), computed as one minus mean rank-biased overlap (RBO) [43], using $p = 0.9$ and retaining the residual tail for ten-word lists. With row-normalised topic–word probabilities β, raw exclusivity E averages $\beta_{kv} / \sum_{j=1}^{K} \beta_{jv}$ over each topic's top ten words and then across topics [3]. Our K-adjustment, $E_{\text{adj}} = (E - 1/K)/(1 - 1/K)$, maps the equal-across-topics baseline $1/K$ to zero. Tables denote this adjusted exclusivity by $\text{EXCL}_{\text{KADJ}}$. Topics with non-finite scores are discarded, and the effective number of topics is reported alongside every score.

5 Results and discussion

5.1 Main comparison and read-outs

On the civil engineering corpus (Table 3), FC-NTM-cTFIDF attains the highest class TF–IDF coherence of all models evaluated ($C_V = 0.636$, NPMI 0.159 over seeds 11/23/37); its generative β coherence is 0.584 (NPMI 0.118), below the NMF β value of 0.598. The raw decoder rows give 0.633 but a much lower exclusivity and are reported in Online Resource 1 (Table S5); we therefore treat class TF–IDF as the primary cross-model read-out and β as a read-out-dependent view. The ten-seed extension preserves the class TF–IDF advantage on the civil corpus but not the β advantage, and Figure 3 shows the per-seed differences behind these means across all three corpora.

Figure 4 compares all evaluated models on the two public corpora under both coherence metrics. On AG News, FC-NTM-cTFIDF reaches $C_V = 0.612$ and NPMI = 0.173, above BTM under the matched class TF–IDF read-out (0.593 / 0.169) and above the β read-outs of ProdLDA, ETM, NMF, CTM-lite and LDA; under its own β

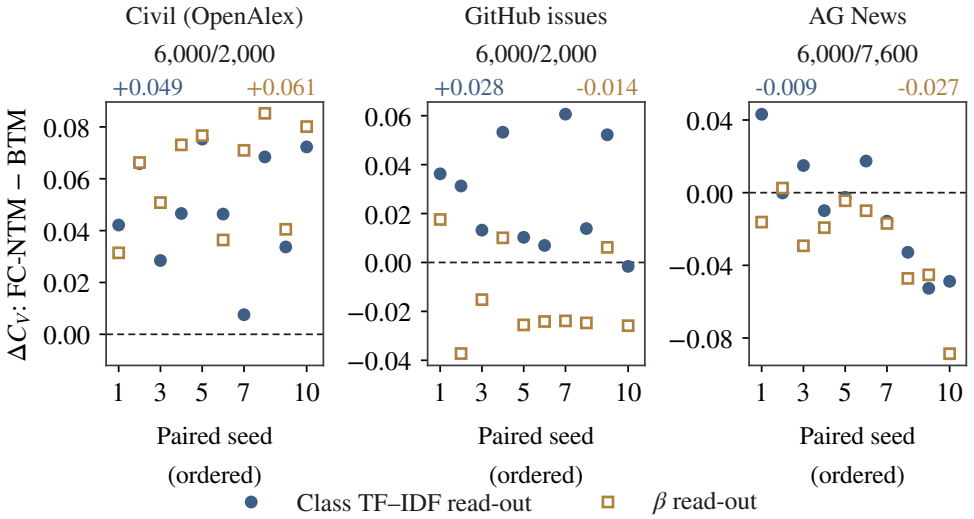


Fig. 3 Per-seed paired comparison of FC-NTM and BTM (ten seeds per corpus, C_V). Each point is one paired seed; the dashed line marks zero difference from BTM. The two values above each panel are mean paired differences (left: class TF–IDF; right: β, generative read-out). The class TF–IDF read-out is positive in all ten paired seeds on the civil corpus and in nine of ten on GitHub issues, whereas the AG News seeds are mixed. This is the corpus-dependent pattern that the mean differences summarise

read-out BTM reaches 0.600 versus 0.541 for FC-NTM. On GitHub issues, FC-NTM-cTFIDF records $C_V = 0.727$ and NPMI = 0.274, above BTM under the class TF–IDF read-out (0.700 / 0.235) but below BTM's β read-out (0.712 / 0.232; FC-NTM 0.625 / 0.192); it is also above the sentence-embedding baselines (CombinedTM 0.643, BERTopic 0.540).

The matched-read-out advantage over BTM is +0.027 C_V and +0.039 NPMI on GitHub and +0.019 / +0.004 on AG News. On the development-free test subset the class TF–IDF ordering is unchanged (GitHub 0.734 versus 0.711, NPMI 0.284 versus 0.248; AG News 0.610 versus 0.590, NPMI 0.172 versus 0.167).

Read-out-matched comparison.

Table 4 collects the read-out-matched comparison across models and read-outs.

The class TF–IDF read-out depends on the model and the corpus: on GitHub it lifts CTM-lite, ProdLDA and LDA by 0.090, 0.087 and 0.068 and leaves ETM and NMF slightly worse, while on AG News it helps every model except BTM. Under the matched read-out, FC-NTM-cTFIDF leads the strongest GitHub baselines (BTM 0.700, CTM-lite 0.675, ETM 0.669, ProdLDA 0.666) by 0.027–0.061 C_V and 0.039–0.059 NPMI, larger than its seed-to-seed standard deviation (0.022); on AG News the advantage over BTM is +0.019 C_V; under the β read-out the AG News means are 0.541 for FC-NTM and 0.600 for BTM. On the civil corpus NMF (0.607) and ETM

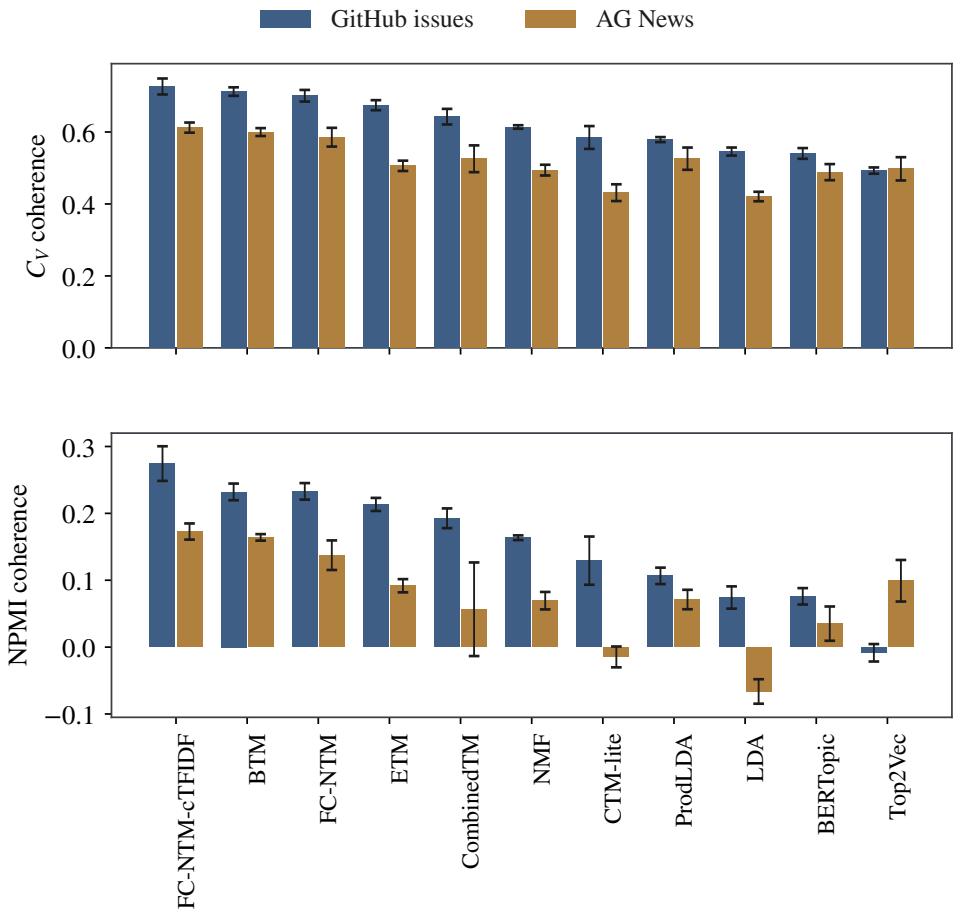


Fig. 4 Main results on the two public corpora (unified evaluation vocabulary, ten words per topic). Top: C_V; bottom: NPMI. The FC-NTM β bars use the generative read-out. Error bars are standard deviations across seeds 11/23/37

(0.603) exceed BTM (0.590), and FC-NTM-cTFIDF leads them by 0.029 and 0.033 C_V. Figure 5 visualises the same comparison.

Ten-seed paired comparison.

Table 5 summarises the ten-seed paired comparisons. We extend FC-NTM and BTM to ten seeds on all three corpora, adding seeds 53, 71, 89, 101, 113, 127 and 139, and compare both read-outs and the occlusion behaviour seed by seed. We compare occlusion with BTM on all three corpora and with LDA on AG News.

For GitHub, the class TF–IDF difference is +0.028 C_V (95% CI $[+0.012, +0.043]$, 9/10 positive seeds), and the relative-degradation difference is +0.024 (10/10). On the civil corpus, the class TF–IDF difference is +0.049 C_V (10/10) and the β difference

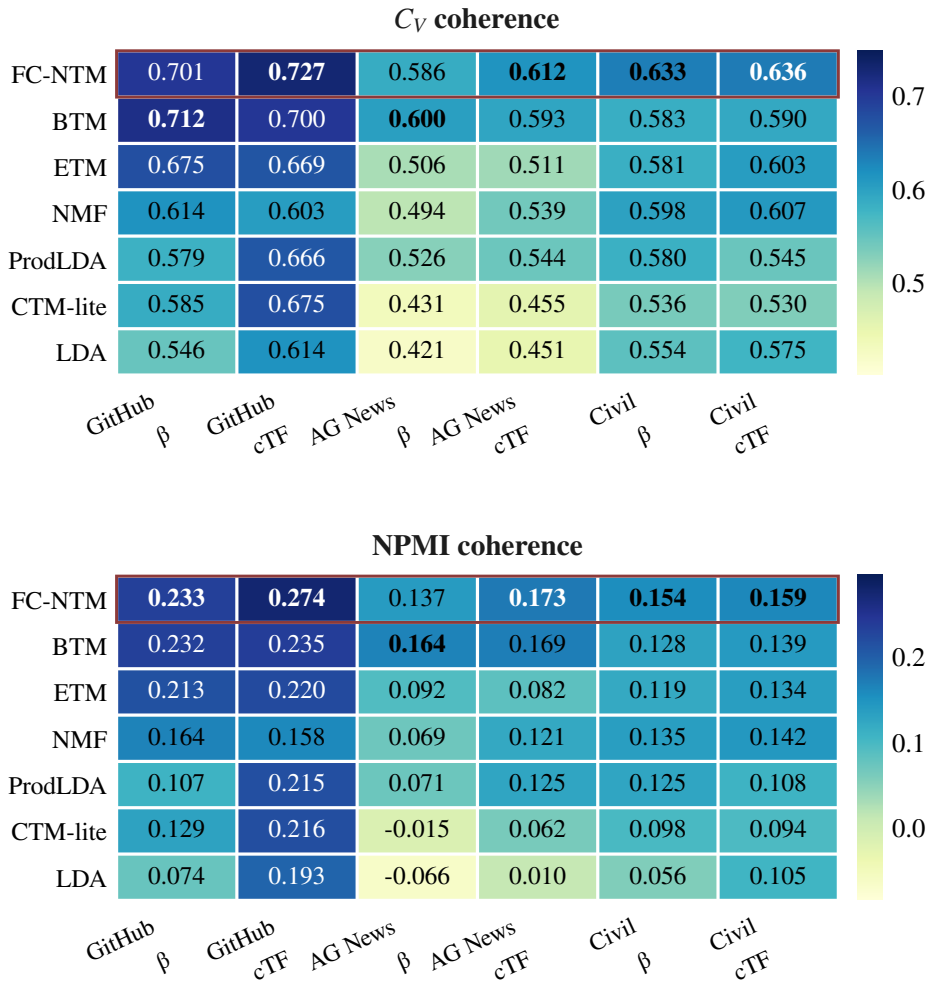


Fig. 5 The read-out comparison of Table 4 as a heatmap ($K = 30$, means over seeds 11/23/37). Top: C_V; bottom: NPMI. Columns combine the three corpora with the two read-outs (β read-out; class TF–IDF, abbreviated cTF). The FC-NTM β columns use the generative read-out. The boxed first row is FC-NTM; bold entries mark the column-wise best

is +0.004 (6/10; 95% CI $[-0.016, +0.023]$); the occlusion difference is −0.003 (0/10). On AG News, the class TF–IDF difference is −0.009 (95% CI $[-0.030, +0.013]$, 3/10) and the β difference is −0.073 (0/10). The β difference on GitHub is −0.094 (0/10). The class TF–IDF mean differences across the three corpora are +0.049, +0.028 and −0.009. The intervals are two-sided 95% t-based intervals.

β **read-out.** Table 5 uses the generative distribution $\mathrm{softmax}(\mathrm{BN}(e_k W))$: mean ten-seed C_V is 0.619 on GitHub, 0.536 on AG News and 0.574 on the civil corpus. The paired difference from BTM is −0.094 on GitHub, −0.073 on AG News and +0.004 on

Table 4 Read-out comparison ($K = 30$, means over seeds 11/23/37). Each model is first read out through its own topic–word matrix (β read-out) and then through a class TF–IDF read-out aggregated from its training-set posterior θ; for FC-NTM the β rows use the generative softmax(BN($e_k W$)) read-out. BTM enters the same read-out through the point-estimate document–topic posterior of its original formulation [47], and the FC-NTM row is the same model as in Tables 1 and 2.

Corpus	Model	β read-out C_V / NPMI	Class TF–IDF C_V / NPMI
GitHub	FC-NTM(3 seed mean)	0.625 / 0.192	**0.727 / 0.274**
	ETM	0.675 / 0.213	0.669 / 0.220
	ProdLDA	0.579 / 0.107	0.666 / 0.215
	CTM-lite	0.585 / 0.129	0.675 / 0.216
	NMF	0.614 / 0.164	0.603 / 0.158
	LDA	0.546 / 0.074	0.614 / 0.193
	BTM(3 seeds)	0.712 / 0.232	0.700 / 0.235
AG News	FC-NTM(3 seed mean)	0.541 / 0.100	**0.612 / 0.173**
	ProdLDA	0.526 / 0.071	0.544 / 0.125
	NMF	0.494 / 0.069	0.539 / 0.121
	ETM	0.506 / 0.092	0.511 / 0.082
	CTM-lite	0.431 / −0.015	0.455 / 0.062
	LDA	0.421 / −0.066	0.451 / 0.010
	BTM(3 seeds)	0.600 / 0.164	0.593 / 0.169
Civil corpus	FC-NTM(3 seed mean)	0.584 / 0.118	**0.636 / 0.159**
	NMF	0.598 / 0.135	0.607 / 0.142
	ETM	0.581 / 0.119	0.603 / 0.134
	BTM(3 seeds)	0.583 / 0.128	0.590 / 0.139
	LDA	0.554 / 0.056	0.575 / 0.105
	ProdLDA	0.580 / 0.125	0.545 / 0.108
	CTM-lite	0.536 / 0.098	0.530 / 0.094

Table 5 Ten-seed paired comparison of FC-NTM and BTM on the three corpora. The read-out rows report C_V differences (FC-NTM minus BTM at the same read-out); the β rows use the generative softmax(BN($e_k W$)) read-out, and the raw decoder view is reported in Online Resource 1 (Table S5). The occlusion rows compare the relative degradation NLL(100%)/NLL(25%), so a positive value means FC-NTM is more stable. Occlusion comparisons use BTM on all three corpora and LDA on AG News. CIs are two-sided 95% t-based confidence intervals; sign and Wilcoxon tests are two-sided.

Corpus	Comparison	Mean diff.	CI low	CI high	Positive	Sign p	Wilcoxon p
GitHub	Class TF–IDF	+0.028	+0.012	+0.043	9/10	0.021	0.004
GitHub	β read-out	−0.094	−0.112	−0.075	0/10	0.002	0.002
GitHub	Occlusion (vs. BTM)	+0.024	+0.022	+0.026	10/10	0.002	0.002
AG News	Class TF–IDF	−0.009	−0.030	+0.013	3/10	0.344	0.375
AG News	β read-out	−0.073	−0.092	−0.054	0/10	0.002	0.002
AG News	Occlusion (vs. BTM)	+0.025	+0.024	+0.026	10/10	0.002	0.002
AG News	Occlusion (vs. LDA)	−0.013	−0.014	−0.012	0/10	0.002	0.002
Civil corpus	Class TF–IDF	+0.049	+0.033	+0.064	10/10	0.002	0.002
Civil corpus	β read-out	+0.004	−0.016	+0.023	6/10	0.754	0.695
Civil corpus	Occlusion (vs. BTM)	−0.003	−0.004	−0.002	0/10	0.002	0.002

the civil corpus (95% CI $[-0.016, +0.023]$, 6/10 positive seeds). The raw decoder rows give higher values (0.701, 0.586 and 0.633 at three seeds) but a much lower exclusivity (0.010–0.014 versus 0.55–0.85); they are reported in Online Resource 1 (Table S5).

Table 6 Field redundancy, measured occlusion gap and the C_V difference against BTM on two corpora. Redundancy is fitted on the training split and evaluated on the test split; the occlusion ΔC_V is the mean±standard deviation over seeds 11/23/37 with dropout training. The last column gives the C_V difference under the matched class TF–IDF read-out and under BTM's β read-out.

Corpus	Held-out Ridge R^2	Held-out ρ^2_{cca}	Occlusion ΔC_V	C_V gap, cTFIDF−BTM (matched / β)
GitHub issues	0.230	0.306	+0.009±0.022	+0.027 / +0.014
AG News	0.158	0.245	+0.018±0.013	+0.019 / +0.012

Cross-model claims use the class TF–IDF read-out; the generative β view is reported separately.

5.2 Field structure and occlusion

Table 6 reports the held-out redundancy measures together with the observed gains, and Figure 6 relates them across the two public corpora.

The redundancy measures are fitted on the training split and evaluated on the test split for the two public corpora. Their held-out predictability is 0.306 for GitHub and 0.245 for AG News (ρ^2_{cca}). The matched class TF–IDF differences from BTM are +0.027 and +0.019; the ten-seed differences are +0.028 (9/10) and −0.009 (3/10). The corresponding civil difference is +0.049 (10/10), as reported in Table 5. Whole-field occlusion changes coherence by $+0.009 \pm 0.022$ on GitHub and $+0.018 \pm 0.013$ on AG News.

Field inference under partial occlusion.

A direct test of field complementarity is whether the visible summary field helps a model recover content when part of the background field is unavailable. The protocol (Online Resource 1) masks a fraction p of the background-field tokens in the test documents, re-infers the topic proportions from the remaining input, and scores the occluded tokens under the model's own word distribution; it reports the per-token negative log-likelihood (NLL), the agreement of the most probable topic with the unoccluded inference, and the posterior JS divergence. At zero visible evidence, BTM and NMF use the training-corpus topic prior. Table 7 reports the per-condition values on GitHub issues, and Figure 7 plots the NLL and relative-degradation curves behind them.

From 25% to 100% occlusion the per-token NLL of FC-NTM rises by 3.2% (7.712 → 7.960), less than LDA (7.4%), ProdLDA (8.0%), ETM (8.9%), BTM (5.2%) and NMF (47.1%), consistent with the training-time whole-field dropout. At full occlusion, BTM has the lowest NLL (6.972), NMF has the highest topic agreement (0.469), followed by ETM (0.452), and FC-NTM reaches 0.358 and a relative degradation of 1.032.

Cross-corpus comparison.

Table 8 and Figure 8 extend the same protocol to the three corpora.

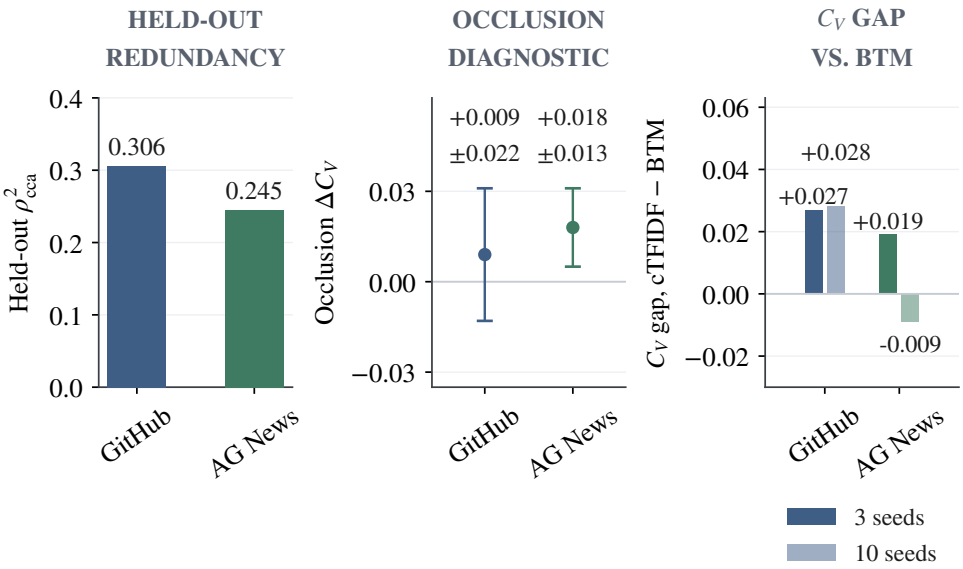


Fig. 6 Field redundancy and gain (held-out measures). Left: squared canonical correlation ρ^2_{cca} (higher means more redundant), fitted on the training split and evaluated on the test split. Middle: occlusion diagnostic ΔC_V (mean±standard deviation over seeds 11/23/37). Right: the C_V difference of FC-NTM-cTFIDF against BTM under the matched class TF–IDF read-out (three seeds) and the ten-seed paired difference. The two corpora reach similar held-out redundancy, while the ten-seed difference is +0.028 on GitHub and −0.009 on AG News

Table 7 Partial-occlusion curves on GitHub issues (seed 11). NLL is the per-token negative log-likelihood of the occluded tokens (lower is better); relative degradation is NLL(100%)/NLL(25%); agreement is the fraction of documents holding their most probable topic at 100% occlusion. BTM enters through its document–topic posterior.

Model	NLL(25%)	NLL(50%)	NLL(75%)	NLL(100%)	Degradation / Agreement
FC-NTM	7.712	7.735	7.790	7.960	**1.032** / 0.358
LDA	6.802	6.854	6.948	7.302	1.074 / 0.420
ProdLDA	7.558	7.588	7.668	8.165	1.080 / 0.354
BTM	6.630	6.639	6.670	**6.972**	1.052 / 0.448
ETM	**6.616**	6.643	6.708	7.202	1.089 / 0.452
NMF	6.972	7.169	7.561	10.255	1.471 / **0.469**

On GitHub (ratio 12.9), all models degrade by factors of 1.028–1.467, and FC-NTM has the smallest relative degradation; the ten-seed paired difference from BTM is $+0.024 \pm 0.004$ (10/10). On the civil corpus (ratio 11.2), the relative degradations are 1.010 for FC-NTM and 1.007 for BTM (paired difference −0.003, 0/10). On AG News (ratio 3.9), they are 1.019 for FC-NTM and 1.006 for LDA. GitHub has the highest

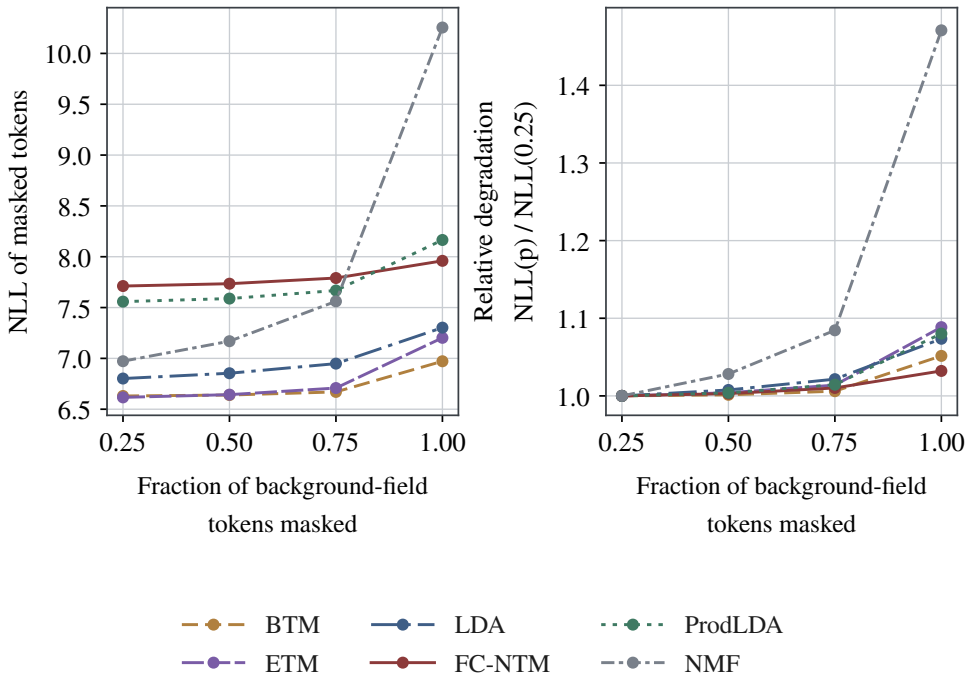


Fig. 7 Inference behaviour under partial occlusion of the background field (GitHub issues, seed 11, 2,000 test documents). Left: per-token negative log-likelihood of the occluded tokens; right: relative degradation NLL(p)/NLL(0.25), which summarises the relative increase from the 25% condition. FC-NTM, trained with whole-field dropout at $p = 0.5$, shows the smallest relative degradation ($\times$1.032). The figure also reports absolute NLL to contextualise this ratio; topic agreement is reported in Table 7

held-out mutual prediction; the length ratio and background-exclusive share separate GitHub from AG News, and the civil corpus falls in a similar range to GitHub.

5.3 Read-out and component analyses

Table 9 compares the field-restricted topic words across models.

Field-restricted read-out truncates the topic words to one field's vocabulary and scores them against that field's own test text. With the shared decoder, the background-view C_V of FC-NTM is 0.623, 0.519 and 0.589 on GitHub, AG News and the civil corpus, and the summary view reaches 0.456, 0.396 and 0.427. BTM is stronger in the background view on GitHub and AG News (0.710 versus 0.623 and 0.586 versus 0.519); NMF leads the civil background view (0.600 versus 0.589); FC-NTM leads the civil summary view (0.427 versus 0.419 for ProdLDA); and LDA and ProdLDA lead the summary view on AG News and GitHub (0.486 and 0.467 versus 0.396 and 0.456).

Table 8 Partial-occlusion behaviour on the three corpora (ten seeds 11–139, mean±standard deviation; the mask is redrawn per seed). Bold marks the smallest relative degradation per corpus. Field asymmetry is described by the mean background/summary token ratio and by the token-type share of background words that do not occur in the summary of the same document.

Corpus(Field asymmetry)	Model	Relative degradation	Agreement(100%)
GitHub issues Length ratio 12.9 Background-exclusive 0.877	FC-NTM	**1.028±0.003**	0.364
	LDA	1.075±0.003	0.416
	ProdLDA	1.078±0.004	0.343
	ETM	1.087±0.003	0.451
	BTM	1.052±0.001	0.443
	NMF	1.467±0.009	0.464
Civil corpus Length ratio 11.2 Background-exclusive 0.894	FC-NTM	1.010±0.001	0.661
	LDA	1.037±0.002	0.574
	ProdLDA	1.028±0.002	0.558
	ETM	1.041±0.001	0.656
	BTM	**1.007±0.001**	0.565
	NMF	1.172±0.002	0.651
AG News Length ratio 3.9 Background-exclusive 0.868	FC-NTM	1.019±0.001	0.515
	LDA	**1.006±0.002**	0.440
	ProdLDA	1.026±0.001	0.491
	ETM	1.072±0.002	0.561
	BTM	1.044±0.001	0.553
	NMF	1.411±0.004	0.545

Fusion reconstruction and gating.

Table 10 separates the effects of fusion reconstruction and learnable gating.

The three-step ablation separates two questions: whether the fused mixture receives its own reconstruction term, and whether fusion is averaged or gated. Adding the fusion-reconstruction term under mean fusion raises the class TF–IDF C_V and NPMI by 0.014 and 0.013 (3/3 seeds) and the generative β read-out by 0.012 and 0.023 (1/3 and 2/3). Replacing mean fusion by a gate leaves the class TF–IDF read-out essentially unchanged but raises the generative β read-out by 0.020 and 0.013 (2/3, 2/3): the gate improves the generative topic matrix without improving the class TF–IDF comparison.

Core-component ablation.

Figure 9 visualises the mean paired differences in Table 11; the table retains the seed-wise positive-pair counts. Each cell is the ablated model minus the full model at the same read-out, so blue indicates lower coherence after removal and red indicates higher coherence. The same zero-centred colour scale is used for all three corpora and both read-outs.

Four components are switched off one at a time. Removing the JS term changes no cell by more than 0.009 C_V. The effect of cross-field reconstruction and word masking depends on the corpus: on GitHub, removing cross-field reconstruction raises the generative β read-out by 0.022 (3/3 seeds) and costs 0.005 in the class TF–IDF read-out, while on AG News and the civil corpus both read-outs drop when either component is removed (cross-field reconstruction $-0.024/-0.004$ and $-0.022/-0.019$;

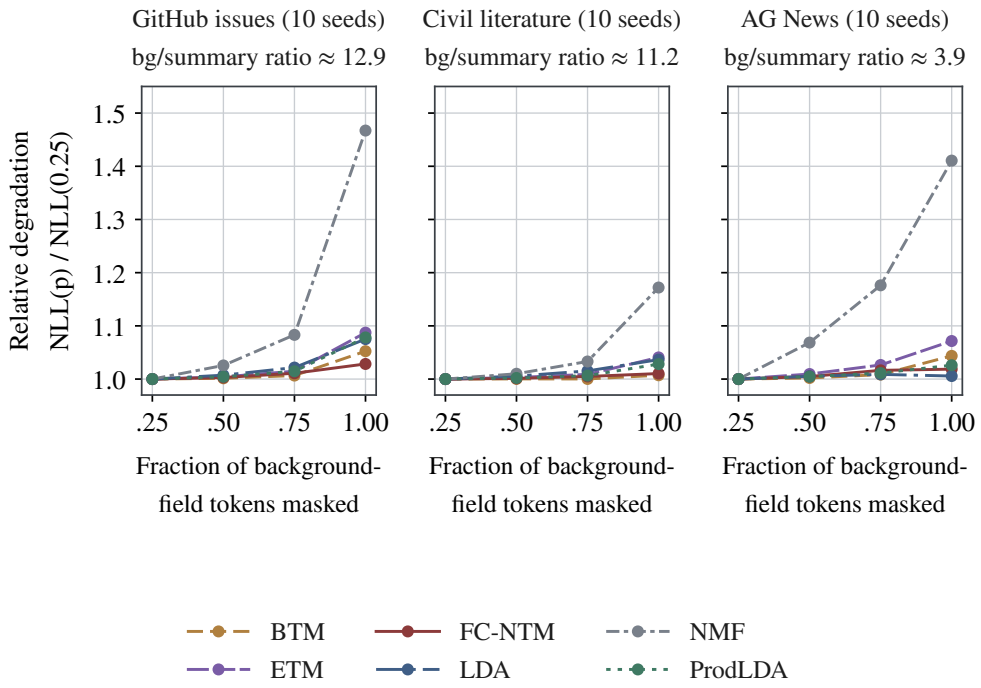


Fig. 8 Relative degradation under the same partial-occlusion protocol on three corpora (curves are means over the ten seeds 11–139 for all six models; endpoint standard deviations are reported in Table 8). Left: GitHub issues (background to summary length ratio ≈ 12.9); middle: civil engineering corpus (≈ 11.2); right: AG News (≈ 3.9). Only on GitHub is the degradation of FC-NTM below all baselines; on the civil corpus and AG News the smallest degradation belongs to BTM and LDA respectively, and the gap to FC-NTM is within 0.003–0.013

word masking $-0.016/-0.011$ and $-0.043/-0.011$; the civil class TF–IDF cells are 0/3). Whole-field dropout lowers both read-outs on GitHub; on AG News it leaves the generative β unchanged (+0.000, 2/3) but lowers the class TF–IDF read-out by 0.023, and on the civil corpus it lowers the β read-out by 0.022 and the class TF–IDF read-out by 0.034.

Sensitivity to the fusion weight.

We checked the sensitivity to the fusion-reconstruction weight directly at the seed level; the text reports the resulting means and paired directions.

At $\lambda_f = 0.25$ the class TF–IDF read-out is essentially the no-fusion result (C_V 0.713 versus 0.714) and the generative β read-out rises by 0.019. Between 0.5 and 1.0 the class TF–IDF gain is stable (0.727 versus 0.723, NPMI 0.274 versus 0.277, differences below 0.005), and the generative β is 0.625 versus 0.621. We use $\lambda_f = 0.5$ throughout.

Table 9 Field-restricted read-out (C_V, mean±standard deviation over seeds 11/23/37). Every topic–word matrix is truncated to the vocabulary of one field and scored against that field's test text only; "background view" is the abstract/body side and "summary view" the title side. FC-NTM's summary view is scored on 20–22 scorable topics per seed on GitHub (29–30 on AG News, 26–28 on the civil corpus) because topics whose ten field words never occur in the held-out text get an undefined C_V. The β entries use the generative softmax(BN($e_k W$)) read-out.

Corpus	Model	Background view C_V	Summary view C_V
GitHub issues	FC-NTM	0.623±0.027	0.456±0.020
	BTM	**0.710±0.011**	0.396±0.018
	NMF	0.609±0.004	0.433±0.002
	LDA	0.543±0.011	0.466±0.005
	ProdLDA	0.575±0.006	**0.467±0.034**
AG News	FC-NTM	0.519±0.036	0.396±0.005
	BTM	**0.586±0.015**	0.399±0.018
	NMF	0.477±0.015	0.444±0.002
	LDA	0.419±0.007	**0.486±0.006**
	ProdLDA	0.499±0.032	0.438±0.007
Civil corpus	FC-NTM	0.589±0.028	**0.427±0.003**
	BTM	0.589±0.014	0.382±0.008
	NMF	**0.600±0.009**	0.398±0.013
	LDA	0.554±0.008	0.415±0.006
	ProdLDA	0.576±0.025	0.419±0.012

5.4 Missing-field inference and controls

Table 12 compares inference with and without the background field. Deleting the background field entirely leaves topic–word coherence within ±0.02 (with sign flips across seeds on GitHub). The information loss appears in the document-level posterior and the word-level prediction: about 66% (GitHub) and 48% (AG News) of the documents change their most probable topic, and the per-token NLL rises by 0.24/0.23 with dropout training (0.24/0.27 without). Whole-field dropout lowers the per-token increase from 0.269 to 0.232 and the posterior JS from 0.060 to 0.049 on AG News; on the civil corpus the values are 0.125→0.103 and JS 0.039→0.033; on GitHub the per-token increase is 0.242 versus 0.241 and the JS improves.

Missing-abstract condition. Masking the abstract of each civil test record and inferring from the title alone leaves the most probable topic unchanged for 67.3% of records (mean JS 0.033); class TF–IDF C_V moves from 0.636 to 0.627 and per-token NLL rises by 0.103. Across seeds 11/23/37, agreement ranges 0.653–0.688 and JS 0.031–0.034. The title-only path remains usable without retraining.

Same-architecture title-only baseline. We also trained FC-NTM itself on titles only (same architecture, 700 epochs, seeds 11/23/37) and scored it on the title-only test reference. Absolute coherence is lower for title-only training, as expected from its shorter training text. On the same test text, the masked full model is +0.010

Table 10 Three-step fusion ablation on GitHub issues ($K = 30$, seeds 11/23/37, same protocol). Within one read-out each row changes exactly one thing: first the fused mixture θ_{fuse} is decoded into the reconstruction term (mean fusion), then mean fusion is replaced by a learnable gate (the main model). "Positive pairs" counts the seeds whose paired difference against the row above is positive (C_V, NPMI). The β entries use the generative softmax(BN($e_k W$)) read-out; the class TF–IDF entries are the primary read-out.

Read-out	Configuration	C_V	NPMI	Positive pairs	UMass
β	No fusion	0.593±0.029	0.156±0.042	—	−5.63
	Mean fusion + fusion recon.	0.605±0.013	0.179±0.019	1/3, 2/3	−5.21
	Gated fusion (main model)	0.625±0.026	0.192±0.038	2/3, 2/3	−5.08
cTFIDF	No fusion	0.714±0.001	0.262±0.007	—	−1.86
	Mean fusion + fusion recon.	0.728±0.007	0.275±0.015	3/3, 3/3	−1.91
	Gated fusion (main model)	0.727±0.022	0.274±0.026	1/3, 1/3	−1.90

Table 11 Core-component ablation on three corpora and two read-outs ($K = 30$, seeds 11/23/37, 700-epoch budget, all other settings identical to the main model). Each column switches off one component: cross-field reconstruction of both streams, the JS consistency weight, field-level word masking, and whole-field dropout; the fusion-reconstruction term is kept in all columns. Values are mean paired ΔC_V (ablated minus full model; negative values indicate lower coherence after removal); parentheses count positive pairs. The β entries use the generative softmax(BN($e_k W$)) read-out; the class TF–IDF entries are the primary read-out.

Corpus	Read-out	−Cross-field reconstruction	−JS consistency	−Word masking	−Field dropout
GitHub	β	+0.022 (3/3)	−0.005 (2/3)	−0.058 (0/3)	−0.017 (1/3)
GitHub	Class TF–IDF	−0.005 (1/3)	+0.004 (2/3)	−0.004 (1/3)	−0.021 (1/3)
AG News	β	−0.024 (1/3)	+0.000 (1/3)	−0.016 (1/3)	+0.000 (2/3)
AG News	Class TF–IDF	−0.022 (0/3)	−0.002 (1/3)	−0.043 (0/3)	−0.023 (1/3)
Civil corpus	β	−0.004 (1/3)	−0.009 (0/3)	−0.011 (1/3)	−0.022 (1/3)
Civil corpus	Class TF–IDF	−0.019 (0/3)	−0.002 (0/3)	−0.011 (0/3)	−0.034 (0/3)

class TF–IDF C_V above the dedicated title-only model (0.451 versus 0.441; 2/3 paired seeds) and +0.019 NPMI above it (0.032 versus 0.013; 3/3), and its generative β is +0.136 higher (0.584 versus 0.448; NPMI 0.118 versus −0.190). The title-only model is higher under the raw decoder read-out (0.428 versus 0.402); the shared test text keeps the comparison paired, while the title-only model is trained on less text.

Title-only-trained baselines. We trained LDA and NMF on titles only and scored them on the same title-only reference text (means over seeds 11/23/37). Under the class TF–IDF read-out, FC-NTM with masked abstracts reaches $C_V = 0.451$ (NPMI 0.032), against 0.427 (NPMI −0.184) for LDA and 0.423 (NPMI −0.093) for NMF. Under the β read-out, LDA reaches 0.456 (NPMI −0.259), FC-NTM 0.402 (NPMI −0.144) and NMF 0.419 (NPMI −0.212). Title-only training does not reproduce the class TF–IDF or NPMI advantage of the full model; the β pattern follows the read-out dependence discussed above.

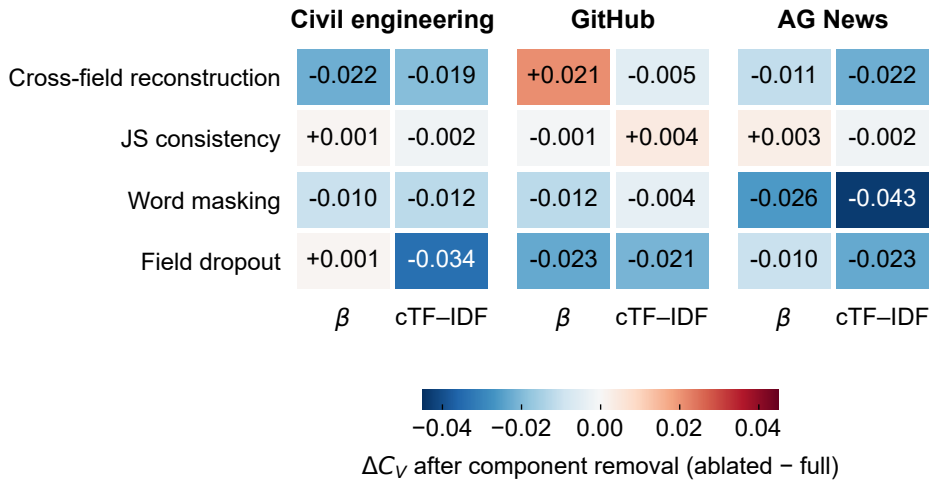


Fig. 9 Core-component ablation effects ($K = 30$, seeds 11/23/37, 700-epoch budget). Cells show mean seed-paired $\Delta C_V = C_V^{\text{ablated}} - C_V^{\text{full}}$; each row removes one component while retaining the fusion-reconstruction term and all other settings. Blue (negative) indicates lower coherence after removal; red (positive) indicates higher coherence. Columns group the civil engineering corpus, GitHub issues and AG News, each with the β and class TF–IDF (cTFIDF) read-outs. The β columns use the generative softmax(BN($e_k W$)) read-out; the cTFIDF columns are the primary read-out. Values are rounded to three decimals. Positive-pair counts are given in Table 11

Table 12 Whole-field occlusion at inference time ($K = 30$, final configuration, 700-epoch budget). ΔC_V is the coherence difference between full-field and occluded inference under the same read-out; ΔNLL is the per-token increase; agreement is the fraction of documents whose most probable topic is unchanged. The two rows of a corpus use the same seeds, isolating the effect of the training-time dropout setting.

Corpus	Training setting	C_V^{full}	C_V^{occ}	ΔC_V	ΔNLL (per token)	Agreement	Mean JS
GitHub	Dropout ($p = 0.5$)	0.727	0.717	+0.009	+0.241	0.345	0.082
	No dropout	0.705	0.725	−0.020	+0.242	0.346	0.092
Civil corpus	Dropout ($p = 0.5$)	0.636	0.627	+0.009	+0.103	0.673	0.033
	No dropout	0.602	0.597	+0.005	+0.125	0.642	0.039
AG News	Dropout ($p = 0.5$)	0.612	0.594	+0.018	+0.232	0.520	0.049
	No dropout	0.589	0.582	+0.007	+0.269	0.498	0.060

Concatenated-field control.

To compare the full method with a single-encoder alternative under the same budget we train a control on the concatenated bag of words $[x_{\text{bg}}; x_{\text{sm}}]$ with the same shared linear decoder, field weights, KL warm-up and orthogonality penalty; whole-field dropout is emulated by zeroing the background half of the input, and there is no fusion or consistency term. The concatenated input doubles the first encoder layer,

keeping the parameter count comparable (a 1.5% difference), and both models use the same budget, vocabulary and read-outs on all three corpora (seeds 11/23/37). Online Resource 1, Table S8 reports the comparison.

Across the six corpus–read-out cells, the mean C_V differences between FC-NTM and the control range from +0.008 to +0.037, all six cells favour FC-NTM, and the paired majorities run from 1/3 to 3/3. The largest gap is on GitHub under the generative β read-out (+0.037), and the civil class TF–IDF gap is +0.029; title-only routing also favours FC-NTM (0.664 versus 0.653 accuracy). The control infers title-only records by zeroing the background half of the input and supports field-restricted read-out with its topic–word matrix.

Title-only routing.

When the abstract of a record is missing, its title can still be routed to an interpretable civil subfield. The evaluation labels are the primary topics that OpenAlex assigns to each record [24]; we use the labels in the frozen snapshot fetched on 12 October 2026. The snapshot, including topic IDs and fetch metadata, is released with the code. The label set is the 15 most frequent topics of the training split, covering 42.9% of the test records (2,000-record test split). Every model infers a topic distribution from the title alone; FC-NTM and the control use their missing-background inference path. The topic-to-label mapping is fitted on the title-only inference: each topic receives the majority training label of the documents assigned to it, with the training-label prior as fallback for topics that receive no training document (three test records in one run). Accuracy and macro-F1 are reported per seed. A supervised TF–IDF logistic regression on the same titles provides a reference point, and FC-NTM given both fields reaches 0.719.

Title-based routing reaches 0.664 accuracy for FC-NTM, 0.653 for the concatenated control, 0.719 with the full fields and 0.810 for the supervised TF–IDF classifier. The corresponding accuracies for the classical topic models are 0.532 for LDA and 0.600 for BTM. Accuracy and macro-F1 for each seed are reported in Online Resource 1, Table S9; Online Resource 1, Table S7 gives four correctly routed title-only examples and two failure cases from the civil test set.

Partial abstract masking.

The title-only condition is the endpoint of a masking process. To measure how the models degrade as the abstract disappears, we mask 10%–100% of the abstract tokens on the civil test split (2,000 records, seeds 53/71/89; the mask is redrawn per seed and rate) and score the topic posterior against full-field inference. Figure 10 reports the mean Jensen–Shannon divergence and top-1 agreement; lower JS means that the full topic distribution is preserved. FC-NTM has the lowest mean JS at every rate: 0.0008 at 10% (next: ProdLDA 0.0016, ETM 0.0029, BTM 0.0129), 0.0072 at 50% (ProdLDA 0.0127, BTM 0.0207, NMF 0.0417), 0.0255 at 80% (ProdLDA 0.0368, BTM 0.0384, NMF 0.0920) and 0.0330 at 100% (BTM 0.1192, ProdLDA 0.1126, LDA 0.1848). Top-1 agreement is comparable to the strongest baselines: at 10% FC-NTM is 0.945 versus NMF 0.947 and ETM 0.944; at 50% it is 0.845 versus NMF 0.852 and ETM 0.854; at 80% it is 0.709 versus NMF 0.765 and ETM 0.768. It remains above BTM at every rate

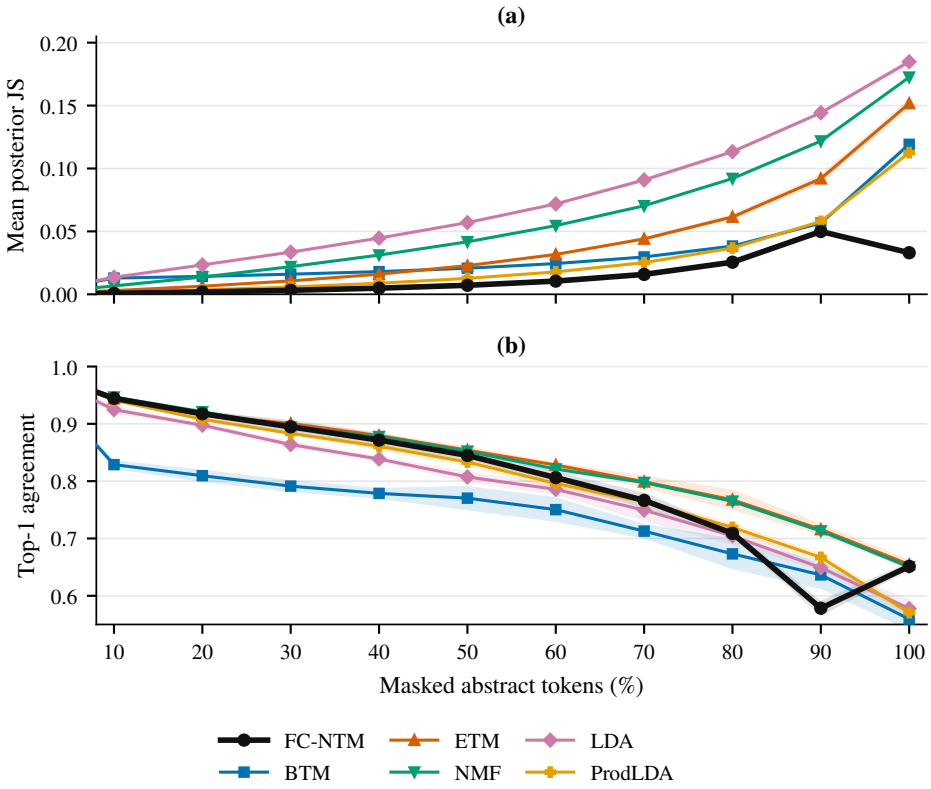


Fig. 10 Partial abstract masking on the civil corpus (2,000 test records, seeds 53/71/89). (a) Mean Jensen–Shannon divergence between the full-field and masked topic posteriors (lower is better); (b) top-1 topic agreement (higher is better). Lines are seed means and bands are ±1 standard deviation over the three seeds; the mask is redrawn per seed and rate. The 100% point is the title-only condition. The two panels report the mean curves: FC-NTM has the lowest posterior JS at every rate and top-1 agreement comparable to the strongest baselines.

from 10% to 80% (0.945 versus 0.829 at 10%; 0.709 versus 0.673 at 80%). Agreement drops to 0.578 at 90% and returns to 0.651 at 100%, where JS is 0.0330. (The 0.651 value is the 100% endpoint of this sweep (seeds 53/71/89); the 67.3% in the abstract and Table 12 is the one-step whole-field occlusion estimate (seeds 11/23/37).) The main effect is posterior stability; agreement varies non-monotonically with the mask rate.

The partial-mask analysis concerns the civil corpus; for the GitHub issues corpus, Figure 11 places the 30 topics on a common scale: bars rank the topics by training-split mass, and each label carries the topic ID and its first three words, complementing the field-restricted comparison in Table 9.

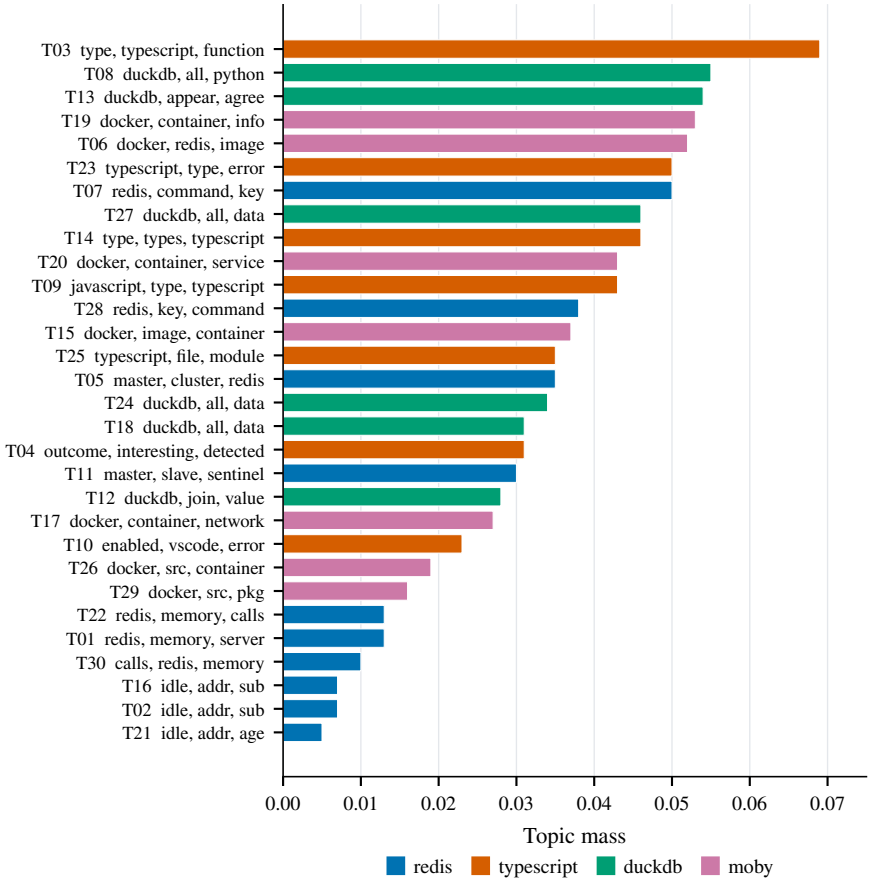


Fig. 11 BERTopic-style topic-mass bar chart for the 30 GitHub topics (FC-NTM, seed 11, $K = 30$). Bars are topic mass in the training split; tick labels give the topic ID and its first three summary-side words. Bar colours correspond to the four source repositories of the GitHub corpus (redis, typescript, duckdb, moby), assigning each topic the repository that dominates its training records. This is a topic-distribution view, not a class TF–IDF weight chart.

6 Discussion and limitations

Across the three corpora, FC-NTM provides a field-aware representation that remains usable when the abstract is absent: one checkpoint supports field-restricted topic words and posterior inference for complete, masked and title-only records. The clearest multi-seed gains are on the civil corpus and the GitHub class TF–IDF read-out; AG News is weaker, with a mean difference of -0.009 and an interval that includes zero. The generative β view differs from the raw decoder rows because generation applies batch normalisation before the softmax; the main β tables use the generative view, and the raw rows are in Online Resource 1 (Table S5). The single-encoder control

recovers most of the mean coherence gain (+0.008 to +0.037), so the two-stream value is functional: separate field posteriors, a trained missing-field path and the consistency diagnostics that the read-out protocol relies on. Masked abstracts retain the most probable topic for 67.3% of civil records (mean JS 0.033), and title-only routing reaches 0.664 accuracy, below the supervised 0.810; the model is therefore a screening tool rather than a routing replacement.

Scope is bounded by three English title/body corpora. Baselines use three seeds against ten for FC-NTM and BTM, and hyper-parameters were selected on development data; the development-free sensitivity analysis removes direct test overlap but not the effect of the initial model selection. The component ablation attenuates rather than eliminates cross-field information flow, and the gate is evaluated with the fusion-reconstruction term; the JS term and global gate add no consistent gain, so the effective configuration is simpler than the full model. The field-restricted summary view averages 20–22 topics for FC-NTM and 30 for the baselines, making those columns descriptive; the title-only comparison is paired but not information-matched. Untested aspects include other languages, field conventions, larger labelled collections, human interpretability and missing-modality VAEs such as JMVAE-kl.

7 Conclusions

This paper presented FC-NTM, a field-cooperative neural topic model for dual-field short texts: two conditional variational streams share one linear topic decoder, cross-field reconstruction and a consistency regulariser encode the field relation into a shared topic space, and field-level word masking and whole-field dropout handle partial and complete field absence. A single trained model supports a decoder β read-out and a class TF–IDF read-out under one evaluation vocabulary, which controls the vocabulary and topic-length confounds of cross-model comparison.

On civil engineering title–abstract records, FC-NTM attains the highest class TF–IDF coherence of the ten methods ($C_V = 0.636$); its generative β coherence is 0.584. It is ahead of BTM in all ten paired seeds on this corpus for class TF–IDF (+0.049 C_V), while the generative β difference is +0.004 (6/10; 95% CI $[-0.016, +0.023]$). The matched-read-out gain is +0.028 C_V on GitHub issues (95% CI $[+0.012, +0.043]$, 9/10 seeds); on AG News the difference is −0.009 (3/10), which bounds the corpus conditions under which the gain appears. A same-budget single-encoder control identifies the specific contribution of the two-stream structure: +0.008–+0.037 coherence together with field-level posteriors and native missing-field inference. Title-only masking retains the most probable topic for 67.3% of civil records, and title-only routing reaches 0.664 accuracy on the 15-topic evaluation subset. The field-relation measurements and occlusion diagnostics separate changes in topic coherence from changes in posterior structure and word-level prediction. The protocol and the per-seed results are released with this paper, and the trained model is available from the authors on reasonable request, so that the framework can be applied to new engineering collections and checked against the same reference results.

Supplementary information

Online Resource 1 contains auxiliary experiments and controls, data and metric definitions, example topics and case studies.

Statements and Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The manuscript sources, figures, supplementary material and per-seed evaluation results used in the reported tables are provided as an anonymized archive accompanying this submission for double-anonymous review. The full model code and trained model checkpoints are available from the authors on reasonable request. A permanent public repository will be linked upon acceptance. Access to the third-party source corpora remains subject to the terms and licences of the original providers. The supplementary material (auxiliary experiments and controls, data and metric notes, example topics and case studies) is provided as Online Resource 1.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large-language-model assistant (OpenAI Codex and a language-model assistant accessed through the DeepSeek API) in order to edit and restructure passages that the authors had drafted, to check numerical consistency between the text, tables and stored result files, and to draft a schematic figure and plotting script that the authors verified against the implementation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The tool was not used to generate research data, to run or select experiments, to produce results, or to draw scientific conclusions, and no figure was generated by a generative image model.

Author contributions

Author contributions and acknowledgements are provided on the separate title page for anonymized review.

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